Chapter 1

Introduction to Operations Strategy

- *Definition and Scope*
- *Strategic Positioning*
- *Levels and Perspectives of Strategy*
- *Relationship between Operations and Competitive Advantage*

Why Do Some Companies Perform Better Than Others?

- *Consider two companies operating in the same industry:*
- *Company A*
  - *Produces high-quality products.*
  - *Delivers orders on time.*
  - *Has relatively high production costs.*
  - *Offers limited product variety.*
- *Company B*
  - *Offers competitive prices.*
  - *Has faster delivery.*
  - *Provides greater product variety.*
  - *Uses more flexible production processes.*

💣*Competitive success depends not only on what a company sells, but also on how it designs and manages its operations.*

What Is an Operation?

- *An operation is a set of activities and processes that transform inputs into outputs.*
- *The Input–Transformation–Output Model*

Inputs	**Transformation Process**	**Outputs**
- Materials - Information - Customers - Facilities and equipment - Human resources	- Manufacturing - Processing - Transportation - Service delivery - Information processing	- Products - Services - Customer experiences - Information

💣 *Every organization has operations, regardless of whether it produces physical goods or services.*

Operations Are Everywhere

- *Operations are not limited to manufacturing companies.*

Industry	*Main operational activities*
Automotive	*Production, assembly, quality control*
Oil and Gas	*Exploration, extraction, refining, distribution*
Electricity	*Generation, transmission, distribution*
Airlines	*Flight operations, maintenance, scheduling*
Hospitals	*Patient admission, diagnosis, treatment*
Banking	*Transaction processing, risk management, customer service*
Retail	*Purchasing, inventory management, distribution*

- *What do these organizations have in common?*

✓ *They all need to coordinate resources, manage processes, satisfy customers, and develop operational capabilities.*

What Is Operations Strategy?

- ***Definition.*** *Operations strategy is the total pattern of decisions that shape the long-term capabilities of an operation and their contribution to overall strategy, through the ongoing reconciliation of market requirements and operations resources.*
- *Four important elements:*
  1. *Pattern of decisions: Strategy is not a single decision; it is a consistent set of related decisions.*
  2. *Long-term orientation: Strategic decisions influence the organization over an extended period.*
  3. *Operational capabilities: Strategy develops the resources and capabilities needed for competitive success.*
  4. *Reconciliation: Strategy connects what the market requires with what operations can deliver.*

Why Is Operations Strategy Important?

- *Operations strategy matters because operational decisions have a direct influence on organizational competitiveness.*
- *It helps organizations to:*
  - *Develop sustainable competitive advantages.*
  - *Improve productivity and resource utilization.*
  - *Reduce operational costs.*
  - *Improve product and service quality.*
  - *Increase delivery speed and reliability.*
  - *Respond to changes in customer demand.*
  - *Develop capabilities that competitors cannot easily replicate.*

💣*A company may have an excellent marketing strategy, but if its production system cannot deliver products on time or maintain quality, its competitive position will eventually deteriorate.*

Operations Strategy vs. Operations Management

- *These two concepts are closely related but have different purposes.*

Operations Strategy	*Operations Management*
Long-term direction	*Day-to-day execution*
Develops operational capabilities	*Uses existing capabilities*
Determines resource configuration	*Coordinates resources*
Determines capacity and technology choices	*Manages production and service processes*
Focuses on future competitiveness	*Focuses on current performance*

- *Example: For an automobile manufacturer:*
  - *Operations strategy: Should the company build a new factory or outsource production?*
  - *Operations management: How many vehicles should the factory produce this week?*

💣*Operations strategy determines the long-term direction; operations management translates that direction into daily activities.*

The Hierarchy of Strategy

- *Operations strategy does not exist independently. It must be consistent with the overall strategy of the organization.*
- *Three levels of strategy:*
  1. *Corporate Strategy*
     - *Which businesses should the organization operate in?*
     - *Which markets and industries should it enter?*
  2. *Business Strategy*
     - *How should the organization compete in a particular market?*
     - *Should it compete through cost, quality, innovation, or differentiation?*
  3. *Functional Strategy*
     - *How should each function support the business strategy?*
     - *What should operations, marketing, finance, and human resources do?*
- *Example: For an automobile manufacturing company:*
  - *Corporate: Enter the electric vehicle market.*
  - *Business: Compete through affordable, reliable vehicles.*
  - *Operations: Develop battery supply, flexible manufacturing, and quality control capabilities.*

Four Perspectives on Operations Strategy

- *There are four perspectives that help us understand how operations strategy develops.*

1. *Top-Down Perspective*
   - *Operations strategy is derived from higher-level business and corporate strategies.*
2. *Bottom-Up Perspective*
   - *Operations strategy develops through experience, learning, and improvement in daily operations.*
3. *Outside-In Perspective*
   - *Operations strategy responds to market requirements, customer expectations, and competitor behavior.*
4. *Inside-Out Perspective*
   - *Operations strategy develops from the resources, knowledge, and capabilities available within the organization.*

Top-Down Perspective

- *Strategy guides operations.*
- *The organization translates its overall strategic objectives into operational decisions.*

- *Strategic hierarchy:*

Corporate Strategy → Business Strategy → Operations Strategy → Operational Decisions

- *Example: Low-Cost Airline*
  - *Business objective: Compete through low ticket prices.*
  - *Operational implications:*
    - *Standardized aircraft fleet.*
    - *High aircraft utilization.*
    - *Fast turnaround times.*
    - *Limited additional services.*
    - *Efficient scheduling and maintenance.*

Bottom-Up Perspective

- *Operational experience shapes strategy.*
- *Operations strategy does not always emerge from top management.*
- *It can also develop through learning and experience.*
- *How does this happen?*
  - *Employees identify operational problems.*
  - *Production teams develop better processes.*
  - *Managers recognize new capabilities.*
  - *Successful improvements become part of the company's competitive strategy.*

- *Example: Toyota Production System*
- *Continuous improvement and learning from production processes have contributed to Toyota's operational capabilities and competitive position.*

💣*Strategy is not only planned at the top; it can also emerge from operational experience.*

Outside-In Perspective

- *Market requirements shape operations strategy.*
- *Organizations must understand what customers value and what competitors offer.*
- *Important considerations:*
  - *Customer expectations*
  - *Market segmentation*
  - *Industry trends*
- *Order Qualifiers: The minimum requirements that a product or service must satisfy to be considered by customers.*
- *Order Winners: The characteristics that make customers choose one product or service over competitors.*
- *Example: Smartphone Market*
  - *Order qualifiers: Basic reliability, acceptable performance, and essential features.*
  - *Order winners: Camera quality, price, design, battery life, or ecosystem.*

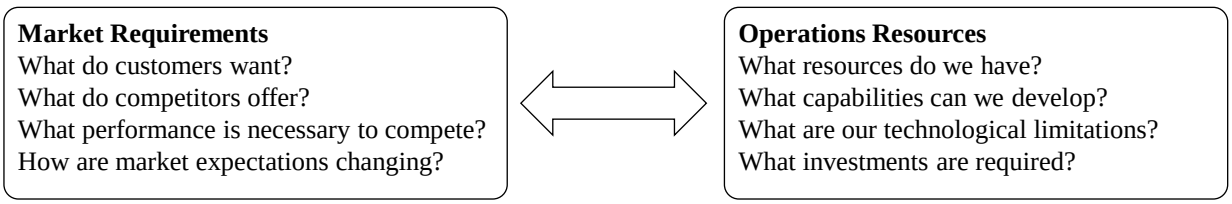
Inside-Out Perspective

- *Operational resources shape competitive strategy.*
- *An organization must understand what it can do particularly well.*
- *Important operational resources:*
  - *Physical facilities*
  - *Production technology*
  - *Human skills and knowledge*
  - *Supply chain relationships*
  - *Management systems*
- *Example: Pharmaceutical Manufacturer*
- *A company with advanced research facilities, specialized technical personnel, and strong quality-control capabilities may develop a competitive advantage in producing complex medicines.*

💣 *What can our organization do exceptionally well, and how can these capabilities create value in the market?*

Reconciling Market Requirements and Operations Resources

- *Organizations must continuously reconcile two dimensions:*



The Strategic Challenge

The organization must develop operational capabilities that satisfy market requirements while making effective use of available resources.

- *Example: Customers demand low prices and rapid delivery, but the company has outdated production technology and high operating costs.*
- *The strategic challenge is to decide how to improve capacity, technology, supply networks, and production processes.*

Strategic Fit and Coherence

- *Operations strategy requires consistency between different levels and areas of decision-making.*
- ***Strategic Fit:** Operational decisions should support the competitive priorities of the business.*
- ***Strategic Coherence:** Different operational decisions should support one another rather than create conflicting objectives.*
- *Example: A company that competes through rapid delivery should consider:*
  - *Appropriate production capacity.*
  - *Reliable suppliers.*
  - *Efficient inventory management.*
  - *Flexible production systems.*
  - *Effective logistics.*
- *Investing in fast delivery while maintaining unreliable suppliers and inflexible production processes creates strategic inconsistency.*

What Decisions Shape Operations Strategy?

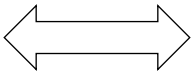
- *Operations strategy involves a range of long-term decisions.*

1. *Capacity Decisions*
   - *How much capacity is required?*
   - *Where should capacity be located?*
   - *When should capacity be expanded?*
2. *Supply Network Decisions*
   - *Which suppliers should be selected?*
   - *What activities should be outsourced?*
   - *Should the company integrate vertically?*
3. *Process Technology Decisions*
   - *Which production technologies should be adopted?*
   - *How much automation is appropriate?*
   - *How should digital technologies be used?*
4. *Development and Improvement Decisions*
   - *How should operational capabilities be developed?*
   - *How can productivity be improved?*
   - *How should products and services be developed?*

Discussion: An Electric Vehicle Manufacturer

- *Consider a company planning to enter the electric vehicle market.*

Market Requirements
Market Requirements
Affordable prices
Reliable batteries
Vehicle safety
Short delivery times
Competitive driving range



Operations Resources
Production capacity and factory location
Battery procurement or in-house production
Manufacturing technology
Supplier relationships
Quality-control systems
Engineering and technical capabilities

- *Strategic Questions*
  1. *Should the company manufacture batteries or purchase them from suppliers?*
  2. *Should it invest in a large factory or begin with smaller production capacity?*
  3. *Should it compete through low prices or product differentiation?*
  4. *Which operational capabilities are most important for long-term competitiveness?*